

AME 3723-001: Numerical Methods for Engineering Computation (Fall 2026)

Mondays, Wednesdays, and Fridays 1:00 – 1:50 PM

Sarkeys Energy Center, Room M204

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Office Hours: Mondays 10:30 AM – 12:00 PM, Wednesdays 10:30 AM – 12:00 PM, or by appointment

TA: Mr. Anish Dhungana

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Office Hours: Thursdays 3:30 PM – 5:00 PM, Fridays 3:30 PM – 5:00 PM

Zoom Link for TA's Office Hours:

<https://oklahoma.zoom.us/j/97809613735?pwd=0uJyJBkIgmraJHAF4Cz8aWnQPDVNOOr.1>

Textbook:

Numerical Methods for Engineers

Steven C. Chapra & Raymond P. Canale, 8th edition, McGraw-Hill, 2021.

Prerequisite:

AME 2402 or C S 1313, and MATH 3113 or MATH 3413

Course Description:

This course introduces essential computational tools engineers use to solve problems that lack closed-form solutions.

Learning Outcomes:

Students will learn error analysis, root finding, systems of equations, optimization, curve fitting, numerical differentiation and integration, as well as numerical solutions to ordinary and partial differential equations.

Topics:

1. Errors inherent in numerical methods (Week 1)
2. Taylor series expansions (Week 1)
3. Root finding techniques (Weeks 2–4)
4. Solution methods for linear algebraic equations (Weeks 5–6)
5. Optimization techniques (Weeks 6–7)
6. Curve fitting techniques (Weeks 9–10)
7. Numerical differentiation and integration techniques (Weeks 10–11)
8. Numerical solutions to ordinary differential equations (Weeks 12–13)
9. Numerical solutions to partial differential equations (Weeks 13–15)

Grade Distribution:

Homework (4).....	20%
Mid-term exam (4).....	50%
Programming project (1)	10%
Final exam (1).....	20%

Grading Policy:

90+	A
80–89	B
70–79	C
60–69	D
–59	F

Canvas:

All lectures, homework assignments, project, solutions, and grades will be posted on Canvas (<https://canvas.ou.edu/>). The students should submit their answers to homework assignments and project to Canvas.

Late Submission and Missed Exam(s):

Unless the instructor gives prior approval, each day (24 hours) of late submission will result in a deduction of 5 points (out of 100 points) for that submission. If your submission is late for more than five days (120 hours), you will get zero point. A missed exam counts as zero and there will be no make-up exam unless a valid excuse from a physician or a university authority is presented to the instructor.

Classroom Policy:

Please refrain from working on homework or other activities during class, and silence your cell phones upon entering. Attendance will not be taken, but active participation is highly encouraged.

Final Exam Preparation Period:

Pre-finals week will be defined as the seven calendar days before the first day of finals. Faculty may cover new course material throughout this week. For specific provisions of the policy please refer to OU's Final Exam Preparation Period policy: <https://www.ou.edu/registrar/academic-records/academic-calendars/final-exam-policies>

Academic Support:

The Gallogly College of Engineering has the infrastructure in place to help you succeed and to get you the support you need. Make sure you take advantage of the infrastructure available to you. For more information, please visit: <https://www.ou.edu/coe/academics/student-support>

Academic Integrity Policy:

Academic integrity means honesty and responsibility in scholarship. Academic assignments exist to help students learn; grades exist to show how fully this goal is attained. Therefore, all work and all grades should result from the student's own understanding and effort. Academic misconduct is any act which improperly affects the evaluation of a student's academic performance or achievement. Misconduct occurs when the student either knows or reasonably should know that the act constitutes academic misconduct. For more information, please visit:

<https://www.ou.edu/integrity/students>

Use of Generative AI in Coursework:

Students may use generative AI tools (e.g., ChatGPT, Gemini, Claude, Copilot, Jasper) to support their learning of course content, homework assignments, and project. However, use of such tools is not permitted during exams.

Reasonable Accommodation Policy:

OU is committed to the goal of achieving equal educational opportunity and full educational participation for students with disabilities. If you have already established reasonable accommodations with the Accessibility and Disability Resource Center (ADRC), please log into <https://blackburn.accessiblelearning.com/OU> to request your semester accommodations as soon as possible and contact the instructor privately, so that we have adequate time to arrange your approved academic accommodations. If you have not yet established services through ADRC, but have a documented disability and require accommodations, please complete ADRC's pre-registration form (https://cm.maxient.com/reportingform.php?UnivofOklahoma&layout_id=350) to begin the registration process. ADRC facilitates the interactive process that establishes reasonable accommodations for students at OU. For more information on ADRC registration procedures, please contact ADRC at (405)325-3852 or adrc@ou.edu, or visit www.ou.edu/adrc for more information. Note: disabilities may include, but are not limited to, mental health, chronic health, physical, vision, hearing, learning and attention disabilities, pregnancy-related. ADRC can also support students experiencing temporary medical conditions.

Religious Observance:

It is the policy of the University to excuse the absences of students that result from religious observances and to reschedule examinations and additional required classwork that may fall on religious holidays, without penalty. [See Faculty Handbook (<https://www.ou.edu/provost/faculty-affairs/faculty-handbook>) 3.15.2]

Adjustments for Pregnancy and Related Issues:

Should you need modifications or adjustments to your course requirements because of pregnancy or pregnancy-related condition, please request modifications via the Institutional Equity Office website (<https://www.ou.edu/eoo/pregnancy-and-parenting>) or call the office at 405/325-3546 as soon as possible. Also, see the Institutional Equity Office FAQ on Pregnant and Parenting Students' Rights (<https://www.ou.edu/content/dam/eoo/documents/faqs/faqs-pregnant-and-parenting-students.pdf>) for answers to commonly asked questions.

Mental Health Support Services:

Support is available for any student experiencing mental health issues that are impacting their academic success. Students can either be seen at the University Counseling Center (UCC) located on the second floor of Goddard Health Center or receive 24/7/365 crisis support from a licensed mental health provider through TimelyCare (<https://www.ou.edu/studentaffairs/resources/timelycare>). To schedule an appointment or receive more information about mental health resources at OU please call the UCC (M–F, 8 AM to 5 PM daily) at 405-325-2911 or visit UCC (<https://www.ou.edu/ucc>), located at 620 Elm Ave., Room 201, Norman, OK 73019.

Title IX Resources and Reporting Requirement:

OU faculty are committed to creating a safe learning environment for all members of our community, free from gender and sex-based discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking, in accordance with Title IX. There are resources available to those impacted, including: speaking with someone confidentially about your options, medical attention, counseling, reporting, academic support, and safety plans. If you have (or someone you know has) experienced any form of sex or gender-based discrimination or violence and wish to speak with someone confidentially, please contact OU Advocates (<https://www.ou.edu/advocacyandeducaon/ou-advocates>, available 24/7 at 405-615-0013) or UCC. Because OU is committed to the safety of you and other students, and because of our Title IX obligations, I, as well as other faculty, Graduate Assistants, and Teaching Assistants, are mandatory reporters. This means that we are obligated to report gender-based violence that has been disclosed to us to the Institutional Equity Office. This means that we are obligated to report gender-based violence that has been disclosed to us to the Institutional Equity Office. This includes disclosures that occur in: class discussion, writing assignments, discussion boards, emails and during Student/Office Hours. You may also choose to report directly to the Institutional Equity Office. After a report is filed, the Title IX Coordinator will reach out to provide resources, support, and information and the reported information will remain private. For more information regarding the University's Title IX Grievance procedures, reporting, or support measures, please visit Institutional Equity Office (<https://www.ou.edu/eoo>) at 405-325-3546.

Emergency Protocol:

During an emergency, there are official university procedures (<https://www.ou.edu/campusafety/policy-and-procedures>) that will maximize your safety.

Severe Weather: If you receive an OU Alert to seek refuge or hear a tornado siren that signals severe weather,

1. Look for severe weather refuge location maps located inside most OU buildings near the entrances.
2. Seek refuge inside a building. Do not leave one building to seek shelter in another building that you deem safer. If outside, get into the nearest building.
3. Go to the building's severe weather refuge location. If you do not know where that is, go to the lowest level possible and seek refuge in an innermost room. Avoid outside doors and windows.
4. Get in, Get down, Cover up
5. Wait for official notice to resume normal activities.

Additional Weather Safety Information is available through the Department of Campus Safety: <https://www.ou.edu/campusafety/divisions#management>

Fire Alarm: If you receive an OU Alert that there is danger inside or near the building, or the fire alarm inside the building activates:

1. LEAVE the building. Do not use the elevators.
2. KNOW at least two building exits
3. ASSIST those that may need help
4. PROCEED to the emergency assembly area

5. ONCE safely outside, NOTIFY first responders of anyone that may still be inside building due to mobility issues.

6. WAIT for official notice before attempting to re-enter the building.

For more information, watch the video for OU Fire Safety on Campus: <https://vimeo.com/125093634>

The University of Oklahoma Active Threat Guidance:

OU embraces a Run, Hide, Fight strategy for active threats on campus. This strategy is well known, widely accepted, and proven to save lives. To receive emergency campus alerts, be sure to update your contact information and preferences in the account settings section at <https://one.ou.edu>.

RUN: Running away from the threat is usually the best option. If it is safe to run, run as far away from the threat as possible. Call 911 when you are in a safe location and let them know from which OU campus you're calling from and location of active threat.

HIDE: If running is not practical, the next best option is to hide. Lock and barricade all doors; turn off all lights; turn down your phone's volume; search for improvised weapons; hide behind solid objects and walls; and hide yourself completely and stay quiet. Remain in place until law enforcement arrives. Be patient and remain hidden.

FIGHT: If you are unable to run or hide, the last best option is to fight. Have one or more improvised weapons with you and be prepared to attack. Attack them when they are least expecting it and hit them where it hurts most: the face (specifically eyes, nose, and ears), the throat, the diaphragm (solar plexus), and the groin.

Please save OUPD's contact information in your phone.

NORMAN campus: For non-emergencies call (405) 325-1717. For emergencies call (405) 325-1911 or dial 911.

TULSA campus: For non-emergencies call (918) 660-3900. For emergencies call (918) 660-3333 or dial 911.

Tentative Semester Schedule:

Week	Date	Topic	Notes	Week	Date	Topic	Notes
1	8/24	Overview		9	10/19	Mid-term exam 2	Cover all chapters since mid-term exam 1
	8/26	Chapter 3	HW 1 assigned, covering Chapters 3 and 4		10/21	Chapter 17	Programming project assigned
	8/28	Chapter 4			10/23	Chapter 18	
2	8/31	Chapter 5		10	10/26	Chapter 18	HW 3 assigned, covering Chapters 17, 18, and 19
	9/2	Chapter 5	HW 1 due		10/28	Chapter 19	
	9/4	No class	OU Football		10/30	Chapter 21	
3	9/7	No class	Labor day	11	11/2	Chapter 21	HW 3 due
	9/9	Chapter 6	HW 1 solution released		11/4	Chapter 22	
	9/11	Chapter 6			11/6	Chapter 23	
4	9/14	Chapter 7		12	11/9	Review	HW 3 solution released
	9/16	Review			11/11	Mid-term exam 3	Cover all chapters since mid-term exam 2
	9/18	Chapter 9			11/13	Chapter 25	
5	9/21	Mid-term exam 1	Cover all chapters to date	13	11/16	Chapter 26	HW 4 assigned, covering Chapters 25, 26, and 27
	9/23	Chapter 9			11/18	Chapter 27	
	9/25	Chapter 10	HW 2 assigned, covering Chapters 9, 10, and 11		11/20	Chapter 29	
6	9/28	Chapter 11		14	11/23	Chapter 30	HW 4 due
	9/30	Chapter 13			11/25	No class	Thanksgiving
	10/2	Chapter 14	HW 2 due		11/27	No class	Thanksgiving
7	10/5	Chapter 14		15	11/30	Chapter 31	HW 4 solution released
	10/7	Chapter 15	HW 2 solution released		12/2	Review	
	10/9	No class	OU holiday		12/4	Mid-term exam 4	Cover all chapters since mid-term exam 3
8	10/12	No class	Instructor's business travel	16	12/7	Review	Programming project due on 12/6
	10/14	No class	Instructor's business travel		12/9	Review	
	10/16	Review			12/11	In-class office hour	

Final exam: December 16, Wed, 10:30 AM – 12:30 PM, in the classroom, will cover all course content.